Data leakage in flight-delay prediction

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How much does including information unavailable two hours before scheduled departure change the measured performance of models predicting arrival delays of at least 15 minutes?

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1. Data and background

Advance flight-delay predictions support travel planning and airline decisions. Wu et al. (2022) evaluated XGBoost models using flight and weather information, identifying departure time as an important predictor. However, strong predictive performance does not necessarily mean a model can support advance decisions. Information known only later, such as actual departure time or later weather observations, may make predictions appear more accurate while being unavailable when needed. Our study investigates how including such information changes model performance, compared with using only information available two hours before scheduled departure, to provide a more practical assessment of flight-delay prediction.

Data sources

- **BTS Reporting Carrier On-Time Performance:** Provides flight information, including flight dates, airlines, origin and destination airports, scheduled departure and arrival times, distance, and arrival-delay outcomes (Bureau of Transportation Statistics, n.d.-a, n.d.-b).
- **Iowa Environmental Mesonet (IEM) METAR/ASOS archive:** Provides airport-level weather observations, including wind speed and precipitation, matched to flight records by location and time (Iowa Environmental Mesonet, n.d.).

Full-study scope

We study completed, non-diverted U.S. domestic flights represented in BTS with observed outcomes, matching weather at origin and destination. January–June 2025 supports training and chronological validation; July– September 2025 is reserved for final testing.

Data Structure

The outcome is `ArrDel15`, indicating arrival at least 15 minutes late, predicted two hours before scheduled departure. Candidate predictors include scheduled departure time and duration, distance, carrier, airports, weekday, weather, holidays, and historical delay rates.

2. Proposed comparison and primary estimands

We will compare logistic regression and random forest under the three information conditions below.

Condition	Weather and operational information
Clean (C)	Information on wind speed and precipitation at the takeoff and landing sites that is available before the cutoff; operational information that is available before the cutoff.
Weather leakage (W)	Includes subsequent wind speed measurements and cumulative precipitation values close to the planned takeoff time based on feature set C.
Weather + operational leakage (WO)	Based on feature set W, further includes unavailable operational information such as `DepDelay`(departure delay) and `TaxiOut`(taxiing time)

- **Primary metric:** Average Precision (AP) summarizes precision across prediction thresholds, weighted by the corresponding increases in sensitivity. It measures how well the model ranks delayed flights above on-time flights and is useful when the outcome classes are imbalanced. *Higher AP indicates better model performance.* (scikit-learn developers, n.d.-a)
- **Secondary metric:** Brier score: The Brier score is the mean squared error between the predicted probability and the actual binary observation. It is introduced to measure the accuracy and calibration of the model's output probability. *A lower Brier score reflects better model performance.* (scikit-learn developers, n.d.-b)

- **The primary estimand is the difference in performance between the leakage conditions and the clean condition.** `Delta_AP = AP_leakage - AP_clean`.

3. Evaluation plan

- **Data Split:** Within the development period, models will be trained on earlier flights and validated on later flights. The final test period will remain untouched until model development is complete.
- **Model Development and Control:** Use only development data for preprocessing fitting (e.g. standardization) and budget-capped hyperparameter tuning. All configurations must be frozen before testing, and clean condition (C) features must be strictly limited to 2 hours before takeoff.
- **Paired comparison:** The clean (C), weather leakage (W), and weather and operational leakage (WO) conditions must be paired and evaluated on the same test flights using equivalent parameter tuning budgets.
- **Evaluation:** Assess AP and Brier score, and report the delay rate and a constant probability reference benchmark based on the development set.
- **Uncertainty:** We will estimate 95% confidence intervals for paired performance differences by resampling test dates, using longer date blocks if needed to account for temporal dependence. Models will remain fixed during resampling.
- **Interpretation:** We will assess the size and uncertainty of performance changes to determine how unavailable information affects measured performance. Conclusions will distinguish apparent gains from improvements usable in advance prediction.

4. Minimum viable experiment

We conducted a preliminary experiment using flights departing from Atlanta in January 2025. Model fitting uses 14,093 flights from January 1 to 20 and the performance was evaluated on 3,636 validation flights from January 21 to 25. The test set contains 4,851 flights from January 26 to 31, which have remained untouched. Weather observations were matched to the corresponding flights at the departure airport.

The preliminary validation results and their interpretation are provided in Appendix A. The completed data matching, model fitting, and evaluation demonstrates the feasibility of our proposed experiment.

The minimum viable experiment compares both models under the three information conditions using a sample limited in time and geographic coverage. The full study will expand coverage to IEM-covered airports across the United States and include weather conditions at destination airports.

5. Computing requirements and verified access to data and code

We will use R and the `ranger` package on a CPU-based laptop (Wright and Ziegler 2017). Data will be downloaded and processed in a month unit.

Access to the required BTS monthly files has been verified, and IEM weather data have been verified for selected airports. The preliminary analysis has confirmed that flight and weather data can be combined for model fitting. Information availability, and computing requirements, including runtime and memory use, will be checked further upon the full study.

6. Fallback and responsibilities

If computing limits require sampling, we will sample proportionally within each month and origin airport. Sample coverage will be checked, and weights will be applied if flights have unequal selection probabilities. Further fallbacks include shortening historical-delay-rate windows from 365 to 182 days and reducing tuning or tree budgets. Forecast data are an option for investigating the association between weather and flight delays.

Responsibilities will be divided as follows:

Name	Responsibilities
Xinnan Zhao	Academic literature review; proposal and final report writing; presentation preparation.
Channing Liu	Data collection and access checks; data cleaning and feature construction; code organization and reproducibility checks.
Coco Liu	Model fitting and tuning; performance evaluation and uncertainty analysis; results interpretation and visualization.

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Appendix A. Preliminary pilot results

Table A1 presents the results of the untuned models on the same 3,636 validation flights departing from Atlanta from January 21 to 25, 2025. The January 26–31 test set was not evaluated.

Table A1. Pilot validation performance under three information conditions.

Model	Clean	Weather leakage only	Weather + operational leakage
Logistic regression	0.2291 / 0.2776	0.2261 / 0.2777	0.9764 / 0.0389
Random forest	0.3509 / 0.2570	0.4286 / 0.2421	0.9710 / 0.0414

Each cell reports AP / Brier, where higher AP and lower Brier indicate better performance.

The weather-leakage condition improved random forest performance but produced little change in logistic regression performance. Adding departure delay and taxi-out duration produced much larger apparent gains for both models.